

○ Taylor Series in two variables

• If $f(x)$ and all its partial derivatives upto n^{th} order are continuous and finite then

$$f(x, y) = f(a, b) + [(x-a)f_x + (y-b)f_y] + \frac{1}{2} [(x-a)^2 f_{xx} + 2(x-a)(y-b)f_{xy} + (y-b)^2 f_{yy}] + \frac{1}{6} [(x-a)^3 f_{xxx} + (y-b)^3 f_{yyy} + 3(x-a)^2(y-b)f_{xxy} + 3(x-a)(y-b)^2 f_{xyy}]$$

★ Calculate $f_x, f_y, f_{xx}, \dots$ at the point (a, b) and put them as coefficients as

Taylor $\equiv$ Polynomials

★ If a point is given, calculate it at the point, If no point then do it on $(0,0)$
(McLaurin Series)

g) expand $e^x \sin y$ upto 3rd degree term

$$f(x, y) = e^x \sin y$$

$$f_x = e^x \sin y = 0$$

$$f_{xxx} = e^x \sin y = 0$$

$$f_{xx} = e^x \sin y = 0$$

$$f(a, 0) = 0$$

$$f_y = e^x \cos y = 1 \quad f_{xy} = -e^x \sin y = 0$$

$$f_{yy} = -e^x \sin y = 0 \quad f_{xxy} = e^x \cos y = 1$$

$$f_{yyy} = -e^x \cos y = -1 \quad f_{xyy} = -e^x \cos y = -1$$

$$e^x \sin y = 0 + [x \cdot 0 + y \cdot (1)] + \frac{1}{2!} [0] + \frac{1}{3!} [x^3(0) + y^3(-1) + 3x^2y(1) + 3xy^2(-1)]$$

$$e^x \sin y = y + \frac{1}{6} [-y^3 + 3x^2y + 3xy^2] + \dots$$

★ g) $e^x \ln(1+y)$, upto 2nd degree

$$f(x, y) = e^x \ln(1+y)$$

$$f(a, 0) = e^0 \ln(1) = (1) \ln(1) = 0$$

$$f_x = e^x \ln(1+y) = 0$$

$$f_{xy} = \frac{e^x}{1+y} = 1$$

$$f_{xx} = e^x \ln(1+y) = 0$$

$$f_y = \frac{e^x}{1+y} = 1$$

$$f_{yy} = \frac{-e^x}{(1+y)^2} = -1$$

$$f(x, y) = 0 + [x \cdot (0) + y \cdot (1)] + \frac{1}{2} [x^2(0) + 2xy(1) + y^2(-1)]$$

$$f(x, y) = y + xy - \frac{y^2}{2} + \dots$$

Partial derivatives aur unki value Neeed Me Mat Nikaalna

• Expanding $f(x, y)$ at a point (a, b)

Kuch oftr, Just use this general form

$$f(x, y) = f(a, b) + [(x-a)f_x + (y-b)f_y] + \frac{1}{2} [(x-a)^2 f_{xx} + (y-b)^2 f_{yy} + 2(x-a)(y-b)f_{xy}] + \dots$$

g) $\sin(xy)$ in powers of $(x-1)$ and $(y-\frac{\pi}{2})$ upto 2nd deg

$$a=1 \quad b=\frac{\pi}{2}$$

$$f(a, b) = \sin \frac{\pi}{2} = 1$$

$$f_x = y \sin xy \quad \cos(xy) y = 0$$

$$f_{xx} = y^2 \sin x - \sin(xy) y^2 = -\frac{\pi^2}{4}$$

$$f_y = \cos(xy) x = 0$$

$$f_{yy} = -\sin(xy) x^2 = -1$$

$$f_{xy} = \cos(xy) + y(-\sin(xy) x) = -\frac{\pi}{2}$$

$$= \cos xy - xy \sin(xy)$$

$$f(x, y) = 1 + [(x-1)0 + (y-\frac{\pi}{2})0] +$$

$$\frac{1}{2} [(x-1)^2 - \frac{\pi^2}{4} + 2xy(-\frac{\pi}{2}) + (y-\frac{\pi}{2})^2]$$

$$f(x, y) = 1 - \frac{1}{2} \left[\frac{\pi^2}{4} (x-1)^2 + \frac{\pi^2}{4} (x-1)(y-\frac{\pi}{2}) + (y-\frac{\pi}{2})^2 \right]$$

9) expand $f(x, y) = x^2y + 3y - 2$ in powers of $(x-1)$ and $(y+2)$

$$a=1 \quad b=-2$$

$$f(a, b) = f(1, -2) = (1)(-2) + 3(-2) - 2 = -2 - 6 - 2 = -10$$

$$f_x = 2xy = 2(1)(-2) = -4$$

$$f_{xx} = 2y = 2(-2) = -4$$

$$f_y = x^2 + 3 = 4$$

$$f_{yy} = 0$$

$$f_{xy} = 2x = 2$$

$$f(x, y) = -10 + [(x-1)(-4) + (y+2)(4)] + \frac{1}{2} [(x-1)^2(-4) + 2(x-1)(y+2)2]$$

$$f(x, y) = -10 - 4(x-1) + 4(y+2) - 2(x-1)^2 + 2(x-1)(y+2) + \dots$$

① Jacobian

Jacobian of (u, v) wrt $(x, y) = J(u, v) = \frac{\partial(u, v)}{\partial(x, y)}$

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$$

Jacobian of (u, v, w) wrt (x, y, z)

$$J = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix}$$

Q1) $u = e^{2x} \sin 3y$ $v = e^{2x} \cos 3y$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2e^{2x} \sin 3y & 3e^{2x} \cos 3y \\ 2e^{2x} \cos 3y & -3e^{2x} \sin 3y \end{vmatrix}$$

$$= -(e^{2x})^2 = -6e^{4x}$$

Q2) $u = 3x - z - 2$
 ~~$y = x$~~ $v = x - 2y + z$
 $w = x^2 + 2xy - zx$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix} = \begin{vmatrix} 3 & 0 & -1 \\ 1 & -2 & 1 \\ (2x+2y-z) & 2x & -x \end{vmatrix}$$

$$= 3(2x - 2x) - 1(2x + 2(2x + 2y - z))$$

$$= -(2x + 4x + 4y - 2z) = -6x - 4y + 2z$$

$$= 2z - (6x + 4y)$$

◎ Maxima and Minima

तुमको $f(x, y)$ दे देगा $\rightarrow$ तुमसे Jaka Joh पुछना है, जी, जी पुछेगा

Step 1 : Calculate $f_x, f_y, f_{xx}, f_{yy}, f_{xy}$

Step 2 : Calculate Stationary points (सक) by

$$f_x = 0 \quad \text{and} \quad f_y = 0$$

Say, we get $x = x_1$ and $y = y_1$

Step 3 : Calculate r, s, t at $x = x_1$ and $y = y_1$

$$r = f_{xx} \quad s = f_{xy} \quad t = f_{yy}$$

$$rt - s^2 > 0 \quad \text{and} \quad r > 0 \rightarrow \text{minima}$$

$$rt - s^2 > 0 \quad \text{and} \quad r < 0 \rightarrow \text{maxima}$$

$$rt - s^2 < 0 \rightarrow \text{NH NM}$$

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xx xy yy

★ Stationary Values :

STANDARD

$$f_x = f(x, y)$$

$$f_y \rightarrow \begin{matrix} x = x_1 \\ y = y_1 \end{matrix}$$

Now put $x = x_1$ in

$$f_x \rightarrow y = y_1, y_2$$

(x_1, y_1) and (x_1, y_2)

put $y = y_1$ in

$$f_x \rightarrow x = x_n$$

$$(x_n, y_1)$$

Trigo $\sin x \sin y \sin(x+y)$ में

Stationary points will be $(0,0)$ and $(\frac{\pi}{3}, \frac{\pi}{3})$